

AI5707 – Pattern Recognition and Machine Learning

Assignment 5: Maximum Likelihood Parameter Estimation

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October 9, 2026

1 Introduction

This assignment studies maximum likelihood estimation (MLE) for three two-dimensional Gaussian classes and uses the estimated parameters for Bayesian classification. The synthetic experiments compare spherical (isotropic), diagonal, and full covariance models, with covariance matrices either shared across classes or estimated separately for each class. A second experiment evaluates a full-covariance Gaussian classifier on the supplied real dataset.

For an observation $\mathbf{x}$ from class ω_i , the Gaussian discriminant is

$$g_i(\mathbf{x}) = -\frac{1}{2} \log |\Sigma_i| - \frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_i)^T \Sigma_i^{-1} (\mathbf{x} - \boldsymbol{\mu}_i) + \log P(\omega_i), \quad (1)$$

and the predicted class is $\hat{y} = \arg \max_i g_i(\mathbf{x})$.

2 Maximum Likelihood Estimation

For class i , with N_i training samples $\mathbf{x}_1, \dots, \mathbf{x}_{N_i}$, the MLEs are

$$\hat{\boldsymbol{\mu}}_i = \frac{1}{N_i} \sum_{n=1}^{N_i} \mathbf{x}_n, \quad \hat{\Sigma}_i = \frac{1}{N_i} \sum_{n=1}^{N_i} (\mathbf{x}_n - \hat{\boldsymbol{\mu}}_i)(\mathbf{x}_n - \hat{\boldsymbol{\mu}}_i)^T. \quad (2)$$

The covariance estimate uses denominator N_i , as required by maximum likelihood estimation.

The three covariance constraints are:

$$\text{Spherical: } \Sigma_i = \sigma_i^2 \mathbf{I}, \quad (3)$$

$$\text{Diagonal: } \Sigma_i = \text{diag}(\sigma_{i1}^2, \sigma_{i2}^2), \quad (4)$$

$$\text{Full: } \Sigma_i = \begin{bmatrix} \sigma_{i1}^2 & \sigma_{i12} \\ \sigma_{i12} & \sigma_{i2}^2 \end{bmatrix}. \quad (5)$$

When all classes share the same covariance matrix, the quadratic terms in pairwise discriminant differences cancel, giving linear decision boundaries. With class-specific covariance matrices, the boundaries are generally quadratic.

3 Part A: Synthetic Gaussian Data

Three classes were generated with 10,000 points per class. Their mean vectors were

$$\boldsymbol{\mu}_1 = [2, 2]^T, \quad \boldsymbol{\mu}_2 = [7, 3.5]^T, \quad \boldsymbol{\mu}_3 = [4, 8]^T.$$

For each covariance configuration, the data were split into 80% training and 20% testing, stratified by class. Thus each experiment used 24,000 training points and 6,000 test points.

3.1 Decision regions

Figure 1 shows the six fitted decision regions with held-out test samples overlaid. The top row uses shared covariance matrices; the bottom row uses class-specific covariance matrices.

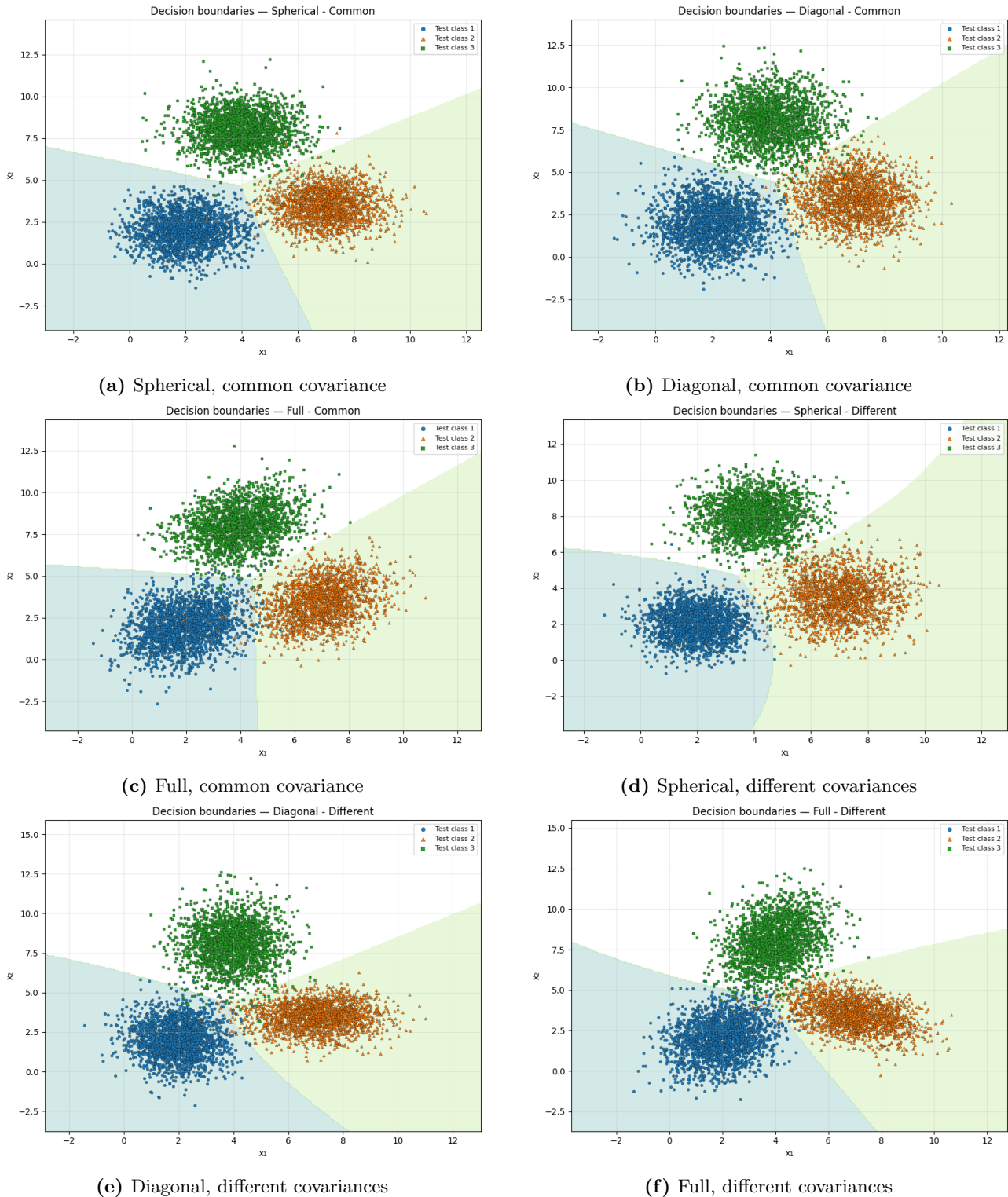


Figure 1: Synthetic-data decision regions and test observations for all six covariance assumptions.

3.2 Accuracy and confusion matrices

Table 1 summarizes the held-out test accuracy. Rows in each confusion matrix are the true classes and columns are the predicted classes.

Table 1: Synthetic test-set classification accuracy.

Covariance	Across classes	Accuracy	Error rate
Spherical	Common	0.9952	0.0048
Diagonal	Common	0.9880	0.0120
Full	Common	0.9877	0.0123
Spherical	Different	0.9903	0.0097
Diagonal	Different	0.9887	0.0113
Full	Different	0.9852	0.0148

Table 2: Confusion matrices for the six synthetic experiments (6,000 test samples per experiment).

A. Spherical, common			
True \ Pred.	ω_1	ω_2	ω_3
ω_1	1996	4	0
ω_2	6	1986	8
ω_3	2	9	1989

Accuracy: 0.9952

C. Full, common			
True \ Pred.	ω_1	ω_2	ω_3
ω_1	1968	17	15
ω_2	25	1972	3
ω_3	10	4	1986

Accuracy: 0.9877

E. Diagonal, different			
True \ Pred.	ω_1	ω_2	ω_3
ω_1	1979	14	7
ω_2	12	1980	8
ω_3	8	19	1973

Accuracy: 0.9887

B. Diagonal, common			
True \ Pred.	ω_1	ω_2	ω_3
ω_1	1981	15	4
ω_2	8	1969	23
ω_3	5	17	1978

Accuracy: 0.9880

D. Spherical, different			
True \ Pred.	ω_1	ω_2	ω_3
ω_1	1985	14	1
ω_2	18	1966	16
ω_3	0	9	1991

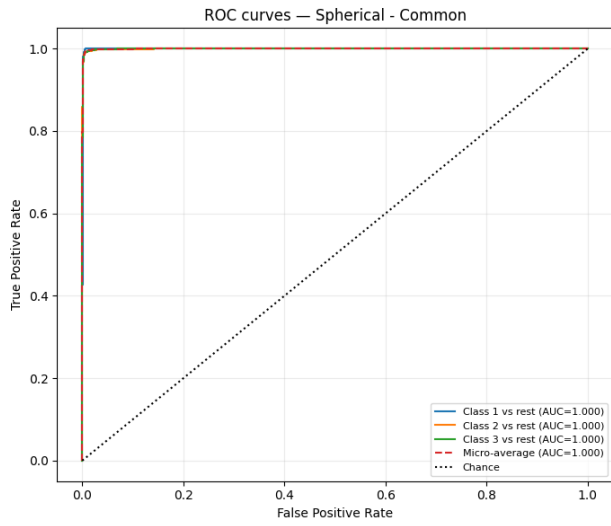
Accuracy: 0.9903

F. Full, different			
True \ Pred.	ω_1	ω_2	ω_3
ω_1	1974	15	11
ω_2	11	1970	19
ω_3	16	17	1967

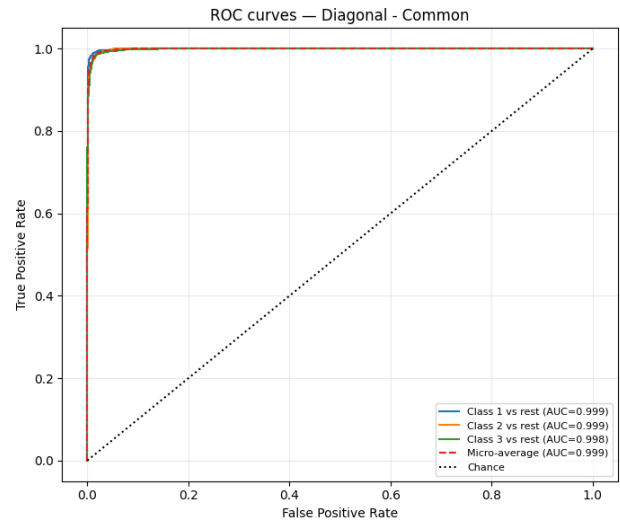
Accuracy: 0.9852

3.3 ROC curves

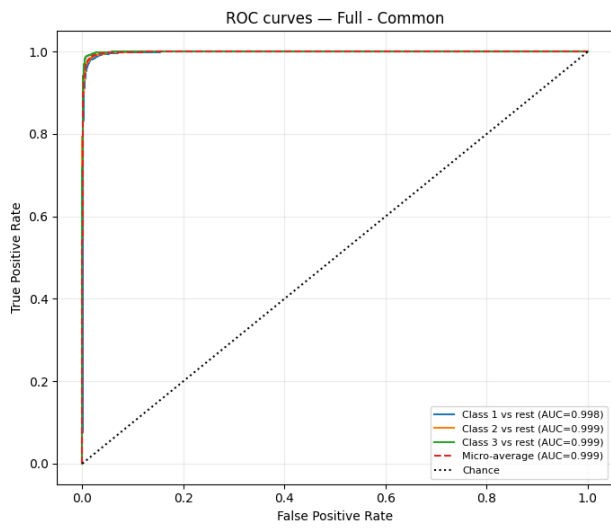
The one-vs-rest ROC curves in Figure 2 evaluate class score rankings across thresholds. AUC values close to 1 indicate strong separation between each class and the remaining classes.



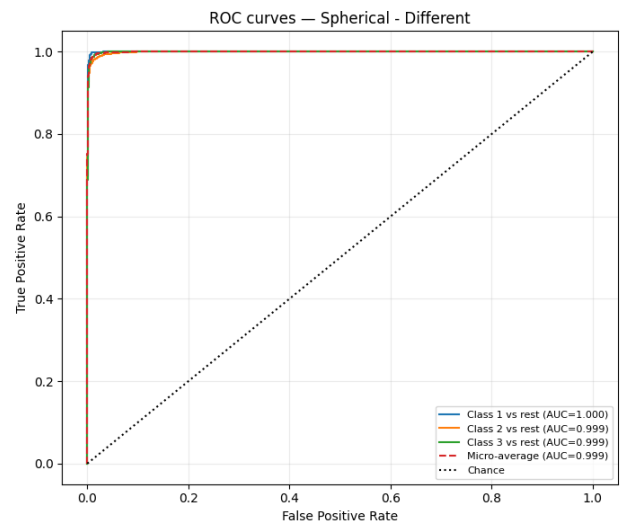
(a) Spherical, common



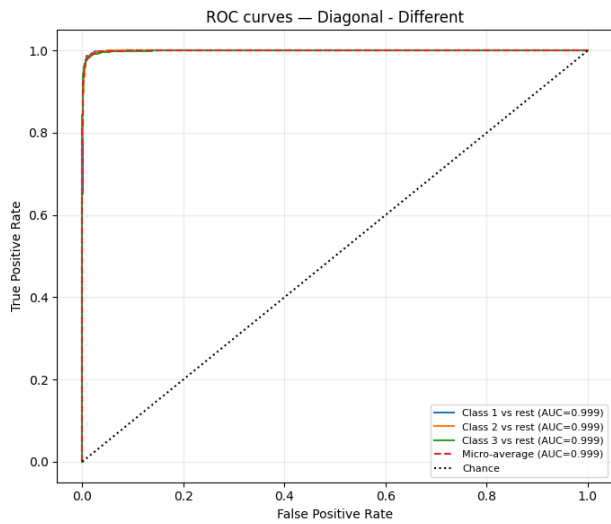
(b) Diagonal, common



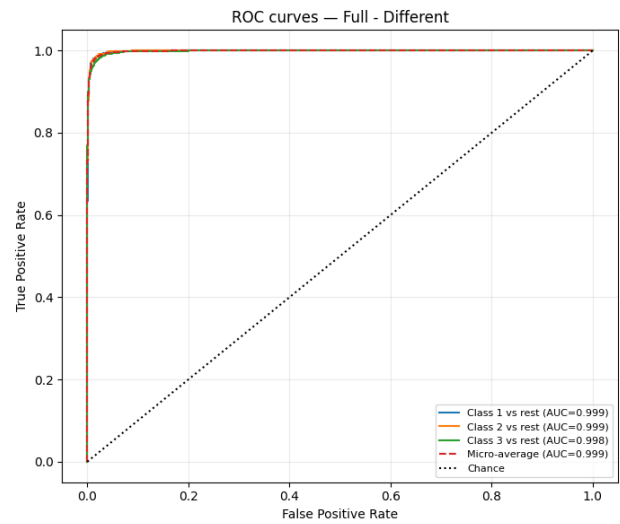
(c) Full, common



(d) Spherical, different



(e) Diagonal, different



(f) Full, different

Figure 2: One-vs-rest ROC curves for the six synthetic classification experiments.

Table 3: Synthetic one-vs-rest AUC values.

Experiment	Class 1	Class 2	Class 3	Micro-average
Spherical, common	0.9997	0.9995	0.9997	0.9996
Diagonal, common	0.9994	0.9988	0.9985	0.9989
Full, common	0.9980	0.9988	0.9994	0.9987
Spherical, different	0.9996	0.9989	0.9993	0.9993
Diagonal, different	0.9993	0.9992	0.9991	0.9992
Full, different	0.9988	0.9987	0.9982	0.9986

4 Part B: Supplied Real Dataset

The provided training file contains 1,050 observations (350 from each class), and the development file contains 300 held-out observations (100 from each class). In this report the provided split is preserved: the training file is used for MLE and the development file is used for testing. Consequently, this section uses the supplied train/development split rather than recombining and randomly re-splitting the two files.

4.1 Training data and estimated parameters

Figure 3 shows the two-dimensional training observations. The estimated class means and full covariance matrices are listed in Table 4.

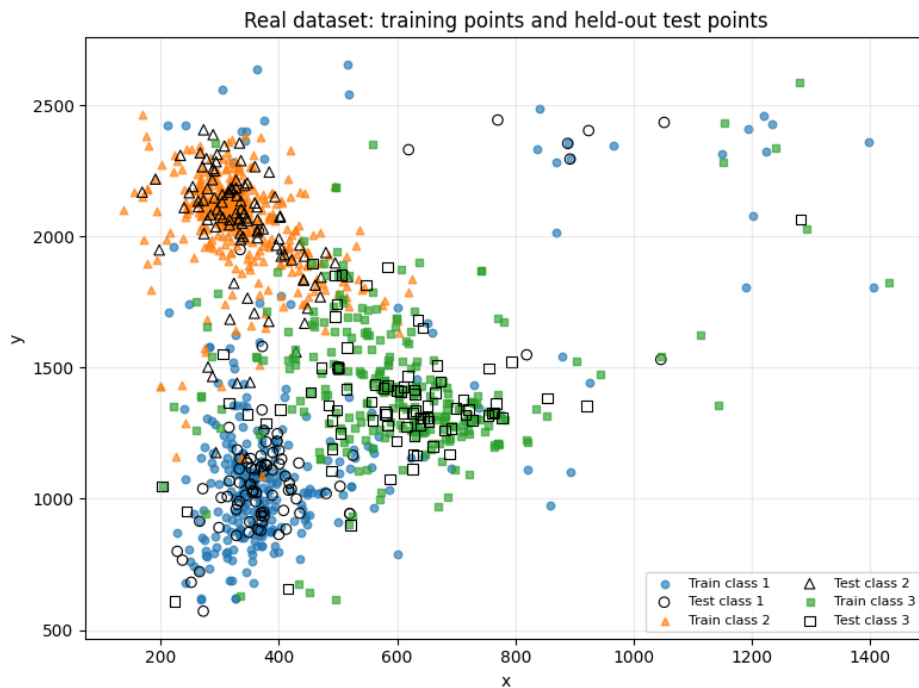
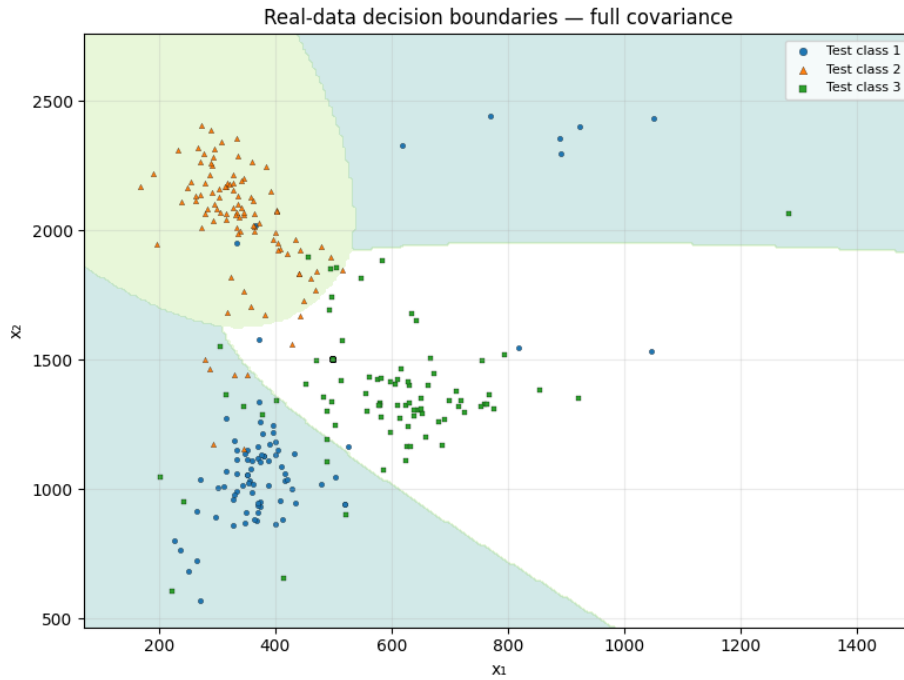
**Figure 3:** Supplied real dataset showing training observations and held-out development observations.

Table 4: Full-covariance MLE estimates from the supplied training data.

Class	Estimated mean $\hat{\boldsymbol{\mu}}_i$	Estimated covariance $\hat{\boldsymbol{\Sigma}}_i$
ω_1	(432.0940, 1241.3891)	$\begin{bmatrix} 33801.0721 & 39788.6866 \\ 39788.6866 & 177814.4940 \end{bmatrix}$
ω_2	(336.6605, 2037.8283)	$\begin{bmatrix} 6783.4304 & -5830.3857 \\ -5830.3857 & 39443.0699 \end{bmatrix}$
ω_3	(575.9242, 1456.2468)	$\begin{bmatrix} 23925.5008 & 3605.1894 \\ 3605.1894 & 59757.2704 \end{bmatrix}$

4.2 Decision boundary and confusion matrix

The full-covariance model achieves 84.33% accuracy on the 300 held-out development observations. Figure 4 overlays these observations on the estimated decision regions. The confusion matrix is shown in Table 5.

**Figure 4:** Decision regions for the real dataset, with the held-out development data overlaid.**Table 5:** Real-data confusion matrix (rows: true class; columns: predicted class).

True \ Predicted	ω_1	ω_2	ω_3	Metric	Value
ω_1	78	2	20	Accuracy	0.8433
ω_2	6	91	3		
ω_3	13	3	84		

4.3 ROC analysis

Figure 5 shows the one-vs-rest ROC curves for the real-data full-covariance classifier. The notebook output reports AUC values of 0.8168, 0.9700, and 0.8775 for classes 1, 2, and 3, respectively, with micro-average AUC 0.9076 (rounded summary table: 0.9077).

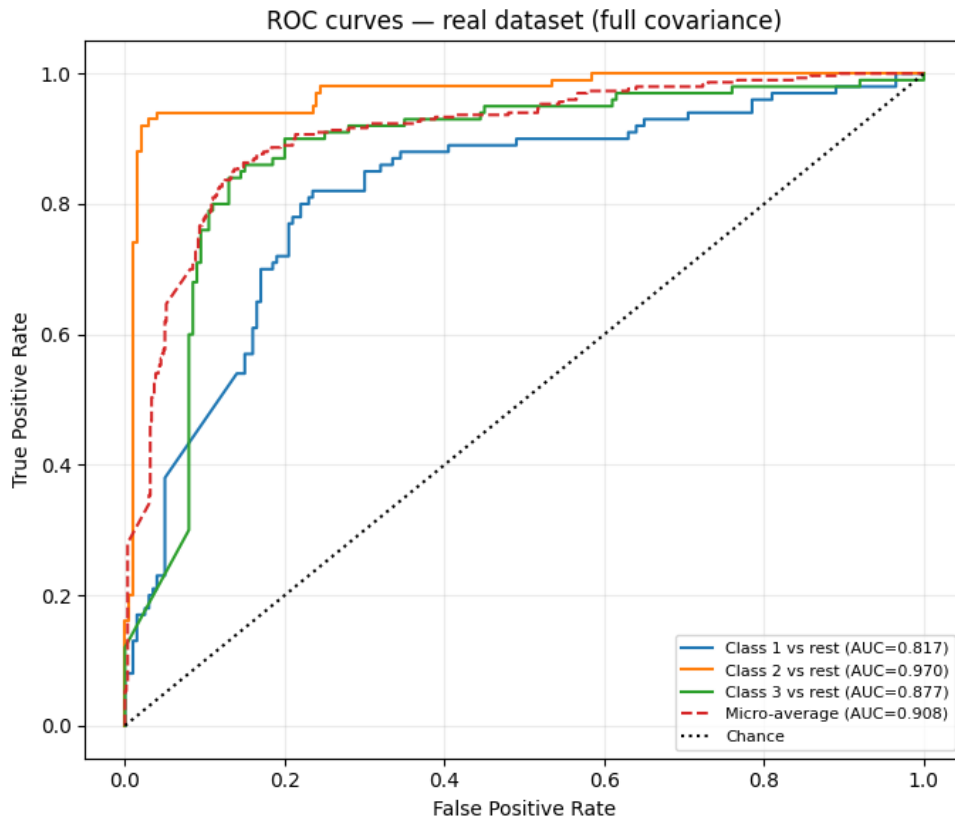


Figure 5: One-vs-rest ROC curves for the real-data full-covariance classifier.

5 Observations

- The synthetic classes are well separated under the selected means and covariance matrices. All six configurations exceeded 98% test accuracy, and the micro-average AUC values are at least 0.9986.
- The spherical common-covariance model has the highest synthetic test accuracy (99.52%) in this run. The result reflects the generated data and the fixed random seed; it should not be interpreted as a universal ranking of covariance models.
- With shared covariance matrices, the Gaussian discriminant boundaries are linear. Allowing different covariance matrices introduces quadratic terms, producing curved boundaries where appropriate.
- On the supplied real dataset, the full-covariance model reaches 84.33% accuracy. Class 2 has the strongest one-vs-rest AUC (0.9700), while class 1 has the lowest (0.8168), indicating that class 1 is more difficult to separate from the other classes by score ranking.
- The real-data confusion matrix shows the largest number of errors for true class 1, of which 20 observations are classified as class 3. This contributes to the lower recall for class 1.

6 Conclusion

Maximum likelihood estimates of the Gaussian means and covariance matrices were used to construct classifiers for the synthetic and supplied real datasets. The synthetic experiments demonstrate how covariance constraints influence decision-region geometry, while the real-data results show that class overlap limits classification performance. Confusion matrices, decision-boundary plots, and ROC/AUC analysis provide complementary views of model performance.